

Performance Analysis of a Nonlinear Energy-Harvesting Cooperative Spectrum-Sharing Network With Statistical Channel Knowledge

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ABSTRACT This paper investigates the performance of an energy-harvesting cooperative spectrum-sharing radio network. The proposed system model includes a pair of primary nodes and a relay that harvests energy from a multi-antenna power beacon, which also operates as a gateway for the relay. The energy beamforming design at the power beacon is based only on statistical channel state information. A realistic nonlinear rectenna model is adopted at the relay to capture both sensitivity and saturation effects, as well as capacitor charging dynamics. Analytical expressions are derived for the outage probabilities of the primary and secondary communication links under Rician fading, and asymptotic analyses are presented for both high- and low-signal-to-noise ratio regimes. The results reveal the presence of outage floors due to energy saturation, as well as reliability trade-offs governed by the energy allocation factor at the relay, nodes distance, and array size. The study provides useful design insights for energy-efficient spectrum-sharing IoT systems envisioned for next generation communication systems.

INDEX TERMS Cooperative communications, energy harvesting, outage probability, spectrum-sharing, statistical channel knowledge, wireless energy transfer.

I. INTRODUCTION

THE Internet of Things (IoT) has emerged as a fundamental technological paradigm, facilitating the interconnection of a massive number of IoT devices and supporting a broad spectrum of applications, ranging from smart city infrastructures to industrial automation systems. Within this paradigm, the concept of massive IoT (mIoT) envisions the large-scale deployment of billions of interconnected IoT devices, each capable of exchanging information about their status and surrounding environment across communication networks. Such an unprecedented scale of deployment significantly intensifies the challenges associated with network sustainability and the longevity of individual devices [1]. A major challenge in realizing mIoT stems from the limited lifespan of battery-powered wireless devices. With a massive

number of deployed nodes, maintaining their energy supply becomes impractical, costly, and environmentally unsustainable. Specifically, the frequent need for battery maintenance disrupts network operation, contributes to environmental pollution through improper disposal and resource depletion, and may also degrade device integrity due to repeated physical interventions, leading to reduced durability and higher maintenance costs [2].

To address this limitation, Energy Harvesting (EH) has emerged as a promising solution to extend the operational lifetime of wireless networks [3], [4]. Several EH techniques have been proposed, including electromagnetic induction, resonant coupling, and harvesting from ambient or dedicated Radio Frequency (RF) signals [5]. Among these, the RF-based Wireless Energy Transfer (WET) approach

stands out due to its long-range energy transfer capability and deployment flexibility [1], [6], which can also be conveniently integrated with Wireless Information Transfer (WIT) [7], [8]. Schemes employing WET for WIT often operate under a harvest-then-transmit principle, wherein devices first perform WET to capture and convert RF signals into Direct Current (DC) power via an integrated rectenna, and subsequently employ WIT to transmit data utilizing the harvested energy.

The efficiency of WET can be further improved through the use of multi-antenna Power Beacons (PBs) that employ distributed energy beamforming. This technique enables the concentration of transmitted energy toward specific spatial directions, thereby maximizing the power delivered to target devices and enhancing the amount of energy harvested by IoT nodes. However, designing the energy beamforming requires Channel State Information (CSI) knowledge at the PB. Although much of the existing literature assumes instantaneous CSI [9], [10], [11], [12], [13], its acquisition introduces substantial signaling and computational overhead, making it impractical for real-world deployments. Consequently, beamforming design based on Statistical CSI (S-CSI) provides a more realistic and energy-efficient solution in practice [14].

Accurate performance analysis of RF-based WET systems requires considering a non-linear EH model [15], [16]. The rectifying antenna (rectenna) inherently exhibits non-linear behavior due to the physical properties of its components, particularly the diode's non-linear response, the presence of a minimum input power threshold, and the saturation of DC output under high RF input levels [17]. Linear models fail to capture these effects accurately, often leading to overly optimistic predictions of harvested energy and system performance. Incorporating a non-linear model thus enables a more faithful representation of the energy conversion efficiency and overall feasibility of WET-enabled communications.

In parallel, the rapid proliferation of IoT has substantially intensified the demand for wireless spectrum resources [18], [19]. This imbalance between device density and available spectrum poses a critical challenge to realize seamless mMTC connectivity. To address this issue, innovative spectrum-sharing paradigms have emerged as promising solutions to enhance spectral efficiency, enable reliable coexistence with primary networks, and ensure the required Quality of Service (QoS) for next-generation IoT applications [18], [20], [21].

In this context, spectrum sharing strategies are primarily classified into two paradigms: underlay and overlay [22]. In the underlay paradigm, secondary and primary systems operate concurrently, provided that interference at the primary receiver remains below a predefined threshold. Conversely, the overlay paradigm allows the secondary node to use prior knowledge of the primary system's messages to cooperate in transmission while simultaneously conveying its own information [23]. Additionally, terminology varies regarding opportunistic access, where secondary users

transmit only during spectral gaps. Some literature classifies this access during idle periods as an overlay strategy [22].

A. RELATED WORKS

The confluence of EH-enabled relays and spectrum-sharing radio networks has drawn considerable attention from the research community [24], [25], [26], [27], [28], [29], [30], [31], [32], [33]. In [24], a cooperative spectrum-sharing architecture employing Non-Orthogonal Multiple Access (NOMA) was analyzed and compared with Orthogonal Multiple Access (OMA) and underlay schemes. The authors derived closed-form outage expressions and showed that, by exploiting NOMA, secondary users can achieve superior spectral efficiency while maintaining the primary user's quality-of-service under interference constraints. Moreover, [25] introduced a full-duplex WET-NOMA cooperative spectrum-sharing framework. In their design, the secondary transmitter simultaneously harvests energy from the primary signal and relays both primary and secondary information. An iterative algorithm for sum-rate maximization was developed, showing substantial gains over half-duplex and OMA-based configurations.

Beyond spectrum-sharing scenarios, several studies have investigated cooperative relay networks powered by RF-based energy harvesting. In [26], a dual-hop half-duplex cooperative IoT network was analyzed, where a decode-and-forward relay harvests energy from received RF signals using a power-splitting-based EH mechanism. Analytical expressions for the average symbol error probability were derived and validated via Monte Carlo simulations, demonstrating that appropriate energy allocation at the relay significantly improves reliability. Similarly, [27] studied a hybrid cooperative wireless system with regenerative and non-regenerative RF energy-harvesting relays under a time-switching EH protocol. By formulating an ergodic capacity optimization problem, the authors showed that hybrid relaying strategies outperform conventional cooperative schemes in terms of fading-averaged capacity.

Considering nonlinear power amplifier effects at the secondary node, [28] proposed a WET-enabled relay model. The study developed an analytical framework capturing amplifier saturation and its impact on harvested energy and overall system throughput. Complementing this line of research, [29] investigated relay selection strategies in nonlinear RF energy-harvesting cooperative networks using a hybrid power-time splitting EH model. Several relay selection schemes based on instantaneous channel conditions were analyzed, and closed-form outage probability and throughput expressions were derived, highlighting the importance of adaptive relay selection under EH constraints. Extending this line of investigation, [30] examined joint subcarrier and power allocation in EH-enabled IoT interference networks. The authors addressed nonlinear EH characteristics by introducing an iterative optimization algorithm that balances

energy transfer efficiency and data transmission reliability. Their results demonstrated that adaptive resource allocation significantly enhances system performance compared with conventional static approaches.

Applying these cooperative EH principles to mission-critical scenarios, [31] analyzed a nonlinear EH-based Industrial IoT network operating under ultra-reliable low-latency communication requirements. The authors derived outage probability and block error rate expressions for both direct and cooperative phases. Their findings emphasized that nonlinear energy-harvesting models are essential for accurately capturing the system performance.

Recent research has increasingly focused on adaptability and energy efficiency in cooperative spectrum-sharing systems. In [32], an adaptive bidirectional transmission strategy was developed for WET-enabled spectrum-sharing relay networks. The proposed method dynamically switches between bidirectional relay and direct transmission modes based on channel and energy conditions, optimizing both spectral and energy efficiency. Analytical and simulation results demonstrated notable improvements in outage probability and throughput relative to fixed transmission strategies.

Most recently, [33] applied deep reinforcement learning to jointly manage time allocation and power control in WET-based spectrum-sharing IoT networks. Modeling the problem as a Markov decision process, the authors employed a double deep Q-network with upper confidence bounds to enable autonomous adaptation to changing radio and energy conditions. The proposed scheme achieved significant gains in long-term throughput and network lifetime under realistic interference and EH constraints.

B. MOTIVATIONS AND CONTRIBUTIONS

Despite these advancements, most existing studies on WET-enabled cooperative spectrum-sharing networks rely on simplified or idealized assumptions. Specifically, many works overlook critical practical implementation challenges, including: (i) the reliance on perfect CSI, assuming the availability of instantaneous CSI for energy beamforming, which is highly impractical due to channel estimation overhead and latency in dynamic environments, as noted in [25], [32] (ii) idealized EH modeling, employing linearized approximations of inherently nonlinear EH circuits or simplified assumptions, leading to inaccurate performance predictions that diverge from real-world behavior, as in [25], [28], [30], [31], and [33] and (iii) focusing on dedicated relay nodes, which limits direct applicability to low-power, autonomous IoT environments where devices must self-sustain, as in [24] and [25].

Furthermore, while NOMA has garnered substantial research attention within wireless communication networks [34], its integration introduces a distinct set of critical issues. NOMA is highly susceptible to inter-user interference and its advertised spectral efficiency gains critically rely on the successful operation of Successive Interference Cancellation (SIC).

Since accurate CSI is crucial for SIC decoding, even minor CSI estimation errors can result in significant performance degradation. The challenge of obtaining perfect CSI is compounded by the growing number of users and the increased device mobility characteristic of modern communication systems [34]. Moreover, obtaining perfect CSI on links involving a node with severe energy limitations, as in the setup discussed in this paper, is rather unrealistic.

Therefore, this paper proposes and rigorously analyzes a novel cooperative spectrum time-sharing framework grounded on practical assumptions to provide mutual reliability benefits for both the Primary Network (PN) and Secondary Network (SN). We investigate a system where an IoT relay assists bidirectional primary communication while opportunistically accessing the spectrum and harvesting energy from a multi-antenna gateway PB. The core novelty lies in concurrently addressing three critical implementation challenges: (i) a topology where the IoT relay facilitates the bidirectional PN link; (ii) the use of S-CSI for multi-antenna energy beamforming, a practical approach that significantly reduces system overhead compared to idealized instantaneous CSI; and (iii) the adoption of a realistic nonlinear EH model that incorporates saturation and capacitor charging dynamics, resulting in a complex yet accurate analysis of the relay's variable transmit power. This focus on the nonlinear model is essential for a robust and practical assessment, as conventional linear models frequently yield optimistic or misleading performance predictions in cooperative and WET networks [35], [36], failing to capture critical effects like saturation and sensitivity thresholds. The overall system performance is rigorously analyzed, including the derivation of asymptotic outage probability equations, to capture the combined impact of these models.

The main contributions of this paper are summarized as follows:

- We propose a cooperative spectrum time-sharing network in which a WET-enabled IoT relay first assists bidirectional primary communication then transmits its own information to a multi-antenna gateway; the latter also acts as PB for the relay. Unlike works assuming instantaneous CSI, the gateway designs its energy beamformer using only S-CSI, a practical assumption for WET.
- We adopt a practical nonlinear EH model with sensitivity and saturation effects and incorporate capacitor charging dynamics, yielding a realistic representation that directly governs the relay's transmit powers across WIT phases.
- We derive outage probability expressions for both the primary and secondary links under Rician fading. For the harvested-power links, closed-form solutions are intractable; hence, the outage is evaluated by conditioning on the received RF power (characterized via Marcum- Q functions) and averaging over its Probability

Density Function (PDF), accounting for S-CSI-based beamforming effects.

- We show that, in the high-Signal-to-Noise Ratio (SNR) regime, EH saturation induces outage floors determined by the WIT link, whereas in the low-SNR regime, insufficient incident power renders the relay ineffective and the primary outage converges to that of the direct link.
- We analyze how the energy-allocation factor at the relay, the gateway-relay distance, the array size, and the time-allocation fractions influence the reliability trade-off between the primary and secondary communications, offering insights into the optimal PB placement and resource partitioning strategies.

For the sake of clarity, Table 1 and Table 2 provide lists of acronyms and symbols. Furthermore, the remainder of this paper is organized as follows. Section II introduces the system model and frame structure. The outage probability analysis is presented in Section III, followed by an asymptotic evaluation in the high and low-SNR regimes in Section IV. Numerical results and performance comparisons are discussed in Section V. Finally, Section VI concludes the paper.

II. SYSTEM MODEL

In this paper, we consider the system model illustrated in Fig. 1. A cooperative overlay spectrum-sharing network is assumed, where two primary nodes, A and B, exchange information with the assistance of a WET-enabled relay node R. The primary and secondary networks do not operate simultaneously, but rather in dedicated, orthogonal time slots. Each primary node can receive signals both directly from the other primary node and indirectly through the relay. The gateway, denoted as G, is located in the proximity of R. All nodes are equipped with a single antenna, except for node G, which is equipped with a Uniform Rectangular Array (URA) comprising M antennas. The URA is configured in the xy -plane with M_X elements along the x -axis and M_Y elements along the y -axis, such that the total number of elements is $M_X \times M_Y$. To maximize the main-lobe gain and minimize grating lobes, the element spacing in both dimensions is set to the standard half-wavelength distance, $d_X = d_Y = \lambda/2$, where $\lambda = c/f_{\text{wet}}$ is the carrier wavelength at the WET frequency. Node G is a dual purpose device, functioning both as a PB and an information gateway for the relay R.

The network operates into multiple orthogonal phases, implemented via Time Division Multiple Access (TDMA), within a time frame of duration T , as illustrated in Fig. 2, in a channel with bandwidth BW. First, during the WET phase, the relay R harvests energy from G, which acts as a PB in this phase and employs its multiple antennas to perform energy beamforming towards R. Then, the PN phase follows. During PN phase 1, node A transmits its data to node B over the direct link and simultaneously to R over the relay link. In PN phase 2, node B transmits its data to node A via the direct link and to R via the relay link. In PN phase 3, the relay node R forwards a combination of the data received in the

TABLE 1. List of Acronyms.

Acronym	Definition
AWGN	Additive White Gaussian Noise
CDF	Cumulative Distribution Function
CSI	Channel State Information
DC	Direct Current
EH	Energy Harvesting
IoT	Internet of Things
LoS	Line-of-Sight
mIoT	massive IoT
MRC	Maximum Ratio Combining
NLoS	Non Line-of-Sight
NOMA	Non-Orthogonal Multiple Access
OMA	Orthogonal Multiple Access
PB	Power Beacon
PDF	Probability Density Function
PN	Primary Network
PS	Power-Splitting
QoS	Quality of Service
RF	Radio Frequency
S-CSI	Statistical CSI
SN	Secondary Network
SNR	Signal-to-Noise Ratio
TDMA	Time Division Multiple Access
TS	Time-Switching
URA	Uniform Rectangular Array
WET	Wireless Energy Transfer
WIT	Wireless Information Transfer

PN phases 1 and 2 to both A and B, employing part of the energy harvested during the WET phase. In the PN phases all nodes transmit with rate r_P . Finally, a SN phase occurs, during which R employs its remaining energy to transmit its own data to G using rate r_S . Moreover, G acts as a gateway in this phase and employs its multiple antennas for receive diversity, improving signal detection reliability. Conversely, during the PN phases, G remains inactive. Specifically, G does not transmit to ensure zero interference to the primary links, nor does it receive, as the primary data is solely intended for nodes A and B.

We denote the durations of the WET phase and the SN phase by $\delta_1 T$ and $\delta_2 T$, respectively, with $\delta_1, \delta_2 > 0$ and $0 < \delta_1 + \delta_2 < 1$. The durations of PN phases 1 to 3 are assumed to be identical, each occupying $(1 - \delta_1 - \delta_2)T/3$. Furthermore, we assume a block fading model in which all channel coefficients remain constant during a single frame and change independently across different frames.

A. WET PHASE

The link between G and R is modeled as a Rician fading channel with Line-of-Sight (LoS) factor κ_S . In the WET phase, G employs energy beamforming to focus energy towards R. The channel vector between G and R is expressed as

$$\mathbf{h}_{GR} = \sqrt{\beta_S} (\bar{\mathbf{h}}_{GR} + \tilde{\mathbf{h}}_{GR}), \quad (1)$$

TABLE 2. List of Symbols.

Symbol	Definition	Symbol	Definition
A, B, R, G	Primary, secondary, relay, and gateway nodes	$p_{\text{out,AB}}, p_{\text{out,AB}}^{\text{direct}}, p_{\text{out,AB}}^{\text{relay}}, p_{\text{out,AB,R}}, p_{\text{out,RG}}$	Outage probabilities of the total, direct, relay-aided, and R-G links
α_P, α_S	Path-loss exponent in the primary and secondary phases	$p_{\text{out,RG}}^{\text{low}}, p_{\text{out,RG}}^{\text{high}}$	Lower and upper bounds of the R-G outage probability
β_{ij}	Average channel power gain between nodes i and j ($i, j \in \{A, B, R\}$)	P_0	Fixed circuitry power
β_S	Average channel power gain of secondary links	P_j	Transmit power of node $j \in \{A, B, R, G\}$
c	Speed of light	$P_{\min}$	Energy harvesting sensitivity
c_0, c_1	Energy-harvesting circuit parameters (rate and turn-on voltage)	$P_{R,\max}$	Maximum transmit power of relay R
C	Capacitance of the storage capacitor	$P_{RG,\max}$	Maximum transmit power of relay R in the secondary phase
d_{ij}	Distance between nodes i and j	P_{RG}	Transmit power of relay R in the secondary phase
d_{GR}	Distance between gateway G and relay R	$P_{S\text{-}CSI}$	RF power harvested by R
δ_1, δ_2	Time fractions of the wireless energy transfer and secondary transmission phases	$P_{th,RG}$	Threshold outage probability for the R-G link
$E_R, E_{R,\max}$	Stored and maximum capacitor energy at relay R	$\Phi(P_{S\text{-}CSI})$	Harvested energy function at R
f	Carrier frequency	r_P, r_S	Primary and secondary phase rates
$f_{P_{S\text{-}CSI}}, F_{P_{S\text{-}CSI}}$	PDF and CDF of the harvested RF power $P_{S\text{-}CSI}$	R_1	Fitting parameter modeling leakage and circuit non-idealities
F_Σ	CDF of the sum of independent random variables	R_{req}	Equivalent series resistance
$\gamma_{ij}, \bar{\gamma}_{ij}$	Instantaneous and average SNR between nodes i and j	$\mathbf{R}_{GR}$	Covariance matrix of the G-R channel
$\gamma_{th,P}, \gamma_{th,RG}$	SNR thresholds for the primary and R-G links	BW	Channel Bandwidth
$\bar{\gamma}_{RB,\max}, \bar{\gamma}_{RG,\max}$	Maximum average SNR for the R-B and R-G links	σ_n^2	Noise variance
$h_{ij}, \bar{h}_{ij}, \tilde{h}_{ij}$	Channel, LoS, and NLoS components between nodes i and j	σ_{RG}^2	Variance of the NLoS components of the R-G link
$h_{RG,m}, \bar{h}_{RG,m}, \tilde{h}_{RG,m}$	Channel, LoS, and NLoS components between R and the m -th antenna of G	$\mathbf{s}_j$	Unit-power transmit signal vector from node $j \in \{A, B, R\}$
$\mathbf{h}_{GR}, \bar{\mathbf{h}}_{GR}, \tilde{\mathbf{h}}_{GR}$	Channel, LoS, and NLoS vectors of the G-R link	$\mathbf{w}_{S\text{-}CSI}$	Precoding vector at gateway G
η	Power amplifier efficiency	W	Energy-harvesting saturation level
κ_P, κ_S	Rician K -factors for the primary and secondary links	$\mathbf{y}_{Aj}, \mathbf{y}_{Bj}, \mathbf{y}_{Rj}$	Received signal at node j from A, B, and R, respectively
M, M_X, M_Y	Total, X-, and Y-axis number of antennas in the URA	$\mathbf{y}_{RG,m}$	Received signal at the m -th antenna of G
μ	Fraction of harvested energy allocated for transmission	$\mathbf{n}_j, \mathbf{n}_{G,m}$	AWGN at node j and the m -th antenna of G
ψ_{m_x}, ϕ_{m_y}	Phase shifts of the (m_x, m_y) -th antenna element in the URA	$\varphi_{X,0}, \varphi_{Y,0}$	Reference phase shifts along the X and Y axes of the URA

where $\mathbf{h}_{GR} \in \mathbb{C}^{1 \times M}$ is the total channel vector, and $\bar{\mathbf{h}}_{GR} \in \mathbb{C}^{1 \times M}$ is given by

$$\bar{\mathbf{h}}_{GR} = \sqrt{\frac{\kappa_S}{1 + \kappa_S}} (\bar{\mathbf{h}}_{GR,Y} \otimes \bar{\mathbf{h}}_{GR,X}), \quad (2)$$

which denotes the deterministic LoS component of the channel vector, with $\otimes$ being the Kronecker product. In addition, $\bar{\mathbf{h}}_{GR,X} \in \mathbb{C}^{1 \times M_X}$ is defined as

$$\bar{\mathbf{h}}_{GR,X} = e^{j\varphi_{X,0}} [1, e^{j\psi_1}, \dots, e^{j\psi_{M_X-1}}], \quad (3)$$

where ψ_{m_x} represents the phase shift of the $(m_x + 1)$ -th antenna element in the X direction of the URA relative to the first antenna, and $\varphi_{X,0}$ is the reference phase shift. Conversely, $\bar{\mathbf{h}}_{GR,Y} \in \mathbb{C}^{1 \times M_Y}$ is defined as

$$\bar{\mathbf{h}}_{GR,Y} = e^{j\varphi_{Y,0}} [1, e^{j\phi_1}, \dots, e^{j\phi_{M_Y-1}}], \quad (4)$$

where ϕ_{m_y} represents the phase shift of the $(m_y + 1)$ -th antenna element in the Y direction of the URA relative to the first antenna with $\varphi_{Y,0}$ as the reference phase shift [37].

Furthermore, $\tilde{\mathbf{h}}_{GR} \in \mathbb{C}^{1 \times M}$ represents the Non Line-of-Sight (NLoS) components of the channel, modeled as a circularly symmetric complex Gaussian random vector $\tilde{\mathbf{h}}_{GR} \sim \sqrt{\frac{1}{1 + \kappa_S}} \mathcal{CN}(\mathbf{0}, \mathbf{R}_{GR})$, where $\mathbf{R}_{GR} = \mathbb{E}[\tilde{\mathbf{h}}_{GR} \tilde{\mathbf{h}}_{GR}^H]$ denotes the covariance matrix. Assuming i.i.d. Rician fading channels,¹ we have $\mathbf{R}_{GR} = \frac{1}{1 + \kappa_S} \mathbf{I}_M$.² In addition, β_S denotes the average channel power and is given by

$$\beta_S = \left(\frac{c}{4\pi f} \right)^2 d_{GR}^{-\alpha_S}, \quad (5)$$

¹The Rician fading model is chosen based on the assumption of partial LoS connectivity. This model provides a robust balance between modeling accuracy, capturing both the deterministic LoS and stochastic multipath components.

²This identity matrix structure for the NLoS covariance implies no correlation among antenna elements. This is justified by the adopted half-wavelength element spacing ($d_X = d_Y = \lambda/2$), which effectively decorrelates scattered components in Rician environments, and aligns with our S-CSI beamforming design that focuses on the dominant LoS component [38], [39].

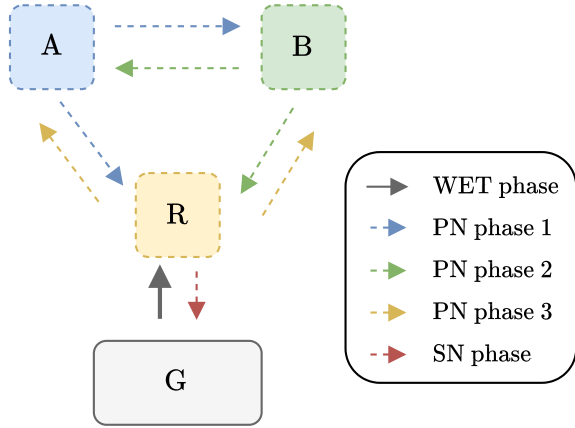


FIGURE 1. System model of the considered cooperative spectrum time-sharing network with WET-enabled relay R and a dual purpose multi-antenna node G, which acts both as a PB and an information gateway for R. The primary nodes A and B communicate with the assistance of R.

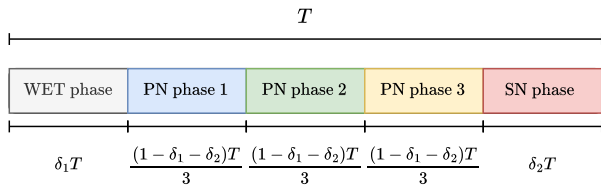


FIGURE 2. Frame structure for the considered cooperative spectrum time-sharing network with WET-enabled relay assistance.

where c is the speed of light, f is the carrier frequency, d_{GR} is the distance between G and R, and α_S is the path loss exponent.

In this paper, the energy beamforming design at the gateway G is based solely on S-CSI knowledge, since acquiring instantaneous CSI is, in general, not practical. Accordingly, the S-CSI based precoding vector $\mathbf{w}_{S\text{-}CSI} \in \mathbb{C}^{1 \times M}$ is [40]

$$\mathbf{w}_{S\text{-}CSI} = \sqrt{P_G} \frac{\bar{\mathbf{h}}_{GR}}{\|\bar{\mathbf{h}}_{GR}\|}, \quad (6)$$

where P_G is the transmit power of G. It is important to note that this beamformer, which relies only on the mean LoS steering vector $\bar{\mathbf{h}}_{GR}$, is sub-optimal for maximizing the average harvested energy. The truly optimal solution would utilize the full channel covariance matrix. However, the LoS-based vector is adopted here due to its low implementation complexity and the assumption of a strong LoS component, which minimizes the performance gap compared to the optimal covariance-based solution.

Thus, neglecting the thermal noise at the relay's receiver for energy harvesting purposes,³ the RF power available for

³We neglect the additive thermal noise at the relay during the WET phase due to the fact that the EH sensitivity threshold is several orders of magnitude larger than the noise power, rendering the noise contribution to the total harvested energy negligible, as commonly assumed in the literature [31], [41], [42].

harvesting by R is

$$\begin{aligned} P_{S\text{-}CSI} &= P_G |\mathbf{h}_{GR} \mathbf{w}_{S\text{-}CSI}^H|^2 \\ &= P_G \beta_S \left| \|\bar{\mathbf{h}}_{GR}\| + \frac{\bar{\mathbf{h}}_{GR}^H \tilde{\mathbf{h}}_{GR}}{\|\bar{\mathbf{h}}_{GR}\|} \right|^2, \end{aligned} \quad (7)$$

which follows a noncentral chi-squared distribution, whose Cumulative Distribution Function (CDF) is given by [43]

$$F_{P_{S\text{-}CSI}}(\xi) = 1 - Q_1\left(\sqrt{2\kappa_S M}, \sqrt{\frac{2\xi(1+\kappa_S)}{P_G \beta_S}}\right), \quad (8)$$

where ξ represents the realization of the random variable $P_{S\text{-}CSI}$, and $Q_1(\cdot, \cdot)$ denotes the first-order Marcum- Q function. The corresponding PDF is given by

$$f_{P_{S\text{-}CSI}}(\xi) = \frac{\theta^2}{2} \exp\left[-\frac{1}{2}(\nu^2 + \theta^2 \xi)\right] I_0(\nu \theta \sqrt{\xi}), \quad (9)$$

for $0 < \xi < W$, where W is the saturation level, $\nu = \sqrt{2\kappa_S M}$, $\theta = \sqrt{\frac{2(1+\kappa_S)}{P_G \beta_S}}$, and $I_0(\cdot)$ represents the modified Bessel function of the first kind.

Then, the harvested energy at R is processed by a rectenna, which converts the received RF power into DC power to charge an energy storage element (e.g., a supercapacitor). The resulting DC power, $\Phi(P_{S\text{-}CSI})$, follows a practical nonlinear model [44]. This model incorporates both a sensitivity threshold $P_{\min}$ and a saturation level W , and is formulated as

$$\Phi(P_{S\text{-}CSI}) = \begin{cases} \frac{W(1 - e^{-c_0 P_{S\text{-}CSI}})}{1 + e^{-c_0(P_{S\text{-}CSI} - c_1)}}, & P_{S\text{-}CSI} > P_{\min}, \\ 0, & \text{otherwise,} \end{cases} \quad (10)$$

where c_0 and c_1 are device-specific constants that define the nonlinear energy harvesting characteristics, corresponding to the harvested energy rate and the minimum turn-on voltage of the EH circuit, respectively. This model is explicitly adopted for its applicability to practical scenarios, as it accurately captures two fundamental physical limitations: the sensitivity threshold $P_{\min}$, below which rectenna diodes fail to activate, and the saturation level W , which caps the output due to circuit limits.

The DC energy stored in the supercapacitor is further constrained by its inherent charging dynamics. In practice, the charging process exhibits nonlinear behavior, implying that the stored energy depends not only on the input power but also on specific circuit parameters and the charging duration. Following the modeling approach in [45], the energy stored in the capacitor can be expressed as

$$E_R = \Phi(P_{S\text{-}CSI}) \delta_1 T \left[1 - e^{-\frac{\delta_1 T}{\tau_C}} \right]^2, \quad (11)$$

where $\tau_C = C(R_{eq} + R_1)$, C is the capacitance, R_{eq} denotes the equivalent series resistance, and R_1 is a fitting parameter that accounts for leakage and other non-idealities.

B. PN PHASES

Assuming the links among the network elements A, B and R have similar propagation characteristics, we model them as Rician fading channels characterized by a LoS factor κ_P . Thus, the channel between the single-antenna nodes $i, j \in \{A, B, R\}$ is expressed as

$$h_{ij} = \sqrt{\beta_{ij}} (\bar{h}_{ij} + \tilde{h}_{ij}), \quad (12)$$

where $\bar{h}_{ij} = \sqrt{\frac{\kappa_P}{1+\kappa_P}} e^{j\phi}$ is the LoS component of the channel, with ϕ being the main phase shift, while $\tilde{h}_{ij} = \sqrt{\frac{1}{1+\kappa_P}} \mathcal{CN}(0, 1)$ denotes the NLoS component of the channel. Furthermore, the average channel power for these links is

$$\beta_{ij} = \left(\frac{c}{4\pi f} \right)^2 d_{ij}^{-\alpha_P}, \quad (13)$$

where d_{ij} is the distance between the nodes i and j , and α_P is the path loss exponent. Moreover, the same values of κ_P and α_P are assumed for PN phases 1, 2, and 3, whose corresponding signal models are presented as follows.

The PN communication proceeds sequentially in three phases: Phase 1 (transmission from node A), Phase 2 (transmission from node B), and Phase 3 (cooperative transmission from relay R). These stages are described by a single generalized signal model, where the signal received at node j from transmitting node k is expressed as

$$\mathbf{y}_{kj} = \sqrt{P_k} h_{kj} \mathbf{s}_k + \mathbf{n}_j, \quad (14)$$

where the transmitter-receiver pairs are defined as: $k = A, j \in \{B, R\}$ for Phase 1; $k = B, j \in \{A, R\}$ for Phase 2; and $k = R, j \in \{A, B\}$ for Phase 3. Here, $\mathbf{n}_j$ denotes the Additive White Gaussian Noise (AWGN) at receiver j with variance⁴ σ_n^2 . The term $\mathbf{s}_k$ represents the transmitted unity-power vector with $\text{BW} \times \delta_1 T$ symbols. Specifically for the relaying phase ($k = R$), the relay forwards the network-coded sequence $\mathbf{s}_R = \mathbf{s}_A \oplus \mathbf{s}_B$, constructed by performing a bit-wise XOR operation on the decoded bit streams from the previous phases. Finally, P_k denotes the transmit power; while P_A and P_B are fixed, the relay transmit power P_R is variable and depends on the energy harvested during the WET phase, as defined in (11).

Let us assume that a fraction μ of the harvested energy E_R is allocated for transmitting the $\mathbf{s}_{\text{XOR}}$ sequence. Considering a fixed circuitry power consumption P_0 and a power amplifier efficiency η , the corresponding available transmit power at the relay in the PN phase 3 is given by

$$P_R = \eta \left[\mu E_R \frac{3}{(1 - \delta_1 - \delta_2)T} - P_0 \right]^+, \quad (15)$$

where $[x]^+ \triangleq \max\{0, x\}$ explicitly enforces the energy causality constraint. The total harvested energy allocated for this phase, μE_R , is first converted into an average supply power by dividing it by the phase duration $\frac{T(1-\delta_1-\delta_2)}{3}$. The fixed circuitry power consumption, P_0 , required for internal operation (e.g., baseband processing) is then subtracted

⁴We assume that the AWGN at all receivers has power σ_n^2 .

from this supply power.⁵ Finally, the result is multiplied by the power amplifier efficiency η , which accounts for the energy loss during the conversion from electrical power to RF transmission power P_R .

C. SN PHASE

Finally, during the SN phase, the relay R transmits its own information to the gateway G by utilizing the remaining portion of its harvested energy, i.e., $(1 - \mu)E_R$, with a corresponding transmit power

$$P_{RG} = \eta \left[(1 - \mu)E_R \frac{1}{\delta_2 T} - P_0 \right]^+. \quad (16)$$

While the expressions in (15) and (16) define the power above the activation threshold, the subsequent outage analysis rigorously accounts for this by treating any realization where the harvested energy is insufficient to cover P_0 as a transmission failure, effectively resulting in an outage event.

The received signal at the m -th antenna of G is given by

$$\mathbf{y}_{RG,m} = \sqrt{P_{RG}} h_{RG,m} \mathbf{s}_R + \mathbf{n}_{G,m}, \quad (17)$$

where $\mathbf{s}_R$ denotes the unity-power symbol vector transmitted by the relay R, $\mathbf{n}_{G,m}$ denotes AWGN, and $h_{RG,m}$ is the channel fading between R and the m -th antenna of G, which, similar to (1), can be written as

$$h_{RG,m} = \sqrt{\beta_S} (\bar{h}_{RG,m} + \tilde{h}_{RG,m}), \quad (18)$$

where $\bar{h}_{RG,m} = \sqrt{\frac{\kappa_S}{1+\kappa_S}} (e^{j\varphi_{X,m}} e^{j\varphi_{Y,m}})$ denotes the deterministic LoS component of the channel vector, with $\varphi_{X,m}$ and $\varphi_{Y,m}$ respectively denoting the antenna phase shift in the X and Y directions of the URA [37]. In addition, $\tilde{h}_{RG,m} \sim \sqrt{\frac{1}{1+\kappa_S}} \mathcal{CN}(0, \sigma_{RG}^2)$ represents the non-LoS component of the channel, with variance $\sigma_{RG}^2 = \frac{1}{1+\kappa_S}$.

III. OUTAGE PROBABILITY ANALYSIS

In this section, we derive the outage probabilities for both the PN and SN communication links. Owing to the practical non-linear EH model, it is analytically intractable to obtain fully closed-form expressions for the outage in the links originating from the relay. Consequently, for the $R \rightarrow A$, $R \rightarrow B$, and $R \rightarrow G$ links, we first evaluate the conditional outage probability for a given amount of harvested power at the relay, and subsequently average it over the distribution of the received RF power, $P_{S\text{-}CSI}$.

A. PN PHASE OUTAGE PROBABILITY

An outage in the PN communication originating from A occurs when both the direct link ($A \rightarrow B$) in PN Phase 1 and the relay-assisted transmission path ($A \rightarrow R \rightarrow B$) fail to achieve the target data rate r_P . Assuming the Shannon limit, the corresponding SNR threshold is $\gamma_{\text{th},P} = 2^{3r_P/(1-\delta_1-\delta_2)} - 1$. Since selection combining is

⁵The dynamic power consumption associated with the decoding process is assumed to be negligible compared to P_0 and the transmission power budget.

assumed between the two received signals at node B, an outage event occurs only when both transmission paths are simultaneously in outage. Therefore, assuming that all wireless channels experience independent fading, the overall outage probability can be expressed as the product of the outage probabilities of the direct and relay-assisted transmission paths:

$$p_{\text{out,AB}} = p_{\text{out,AB}}^{\text{direct}} \times p_{\text{out,AB}}^{\text{relay}}. \quad (19)$$

1) DIRECT PATH OUTAGE PROBABILITY

An outage probability in the direct $A \rightarrow B$ link occurs if $\gamma_{AB} < \gamma_{\text{th,P}}$, whose probability, by means of (8), is given by

$$p_{\text{out,AB}}^{\text{direct}} = 1 - Q_1\left(\sqrt{2\kappa_P}, \sqrt{\frac{2(1+\kappa_P)\gamma_{\text{th,P}}}{\bar{\gamma}_{AB}}}\right), \quad (20)$$

where $\bar{\gamma}_{AB} = P_A\beta_{AB}/\sigma_n^2$ is the average SNR of the $A \rightarrow B$ link.

2) RELAY PATH OUTAGE PROBABILITY

The relay-assisted transmission path $A \rightarrow R \rightarrow B$ is considered successful only if both the $A \rightarrow R$ link (during PN Phase 1) and the $R \rightarrow B$ link (during PN Phase 3) are not in outage. Consequently, the outage probability associated with this relay-assisted path is

$$\begin{aligned} p_{\text{out,ARB}}^{\text{relay}} &= \Pr\{\min(\gamma_{AR}, \gamma_{RB}) < \gamma_{\text{th,P}}\} \\ &= 1 - \Pr\{\gamma_{AR} \geq \gamma_{\text{th,P}}\} \Pr\{\gamma_{RB} \geq \gamma_{\text{th,P}}\}. \end{aligned} \quad (21)$$

The outage probability of the $A \rightarrow R$ link is

$$\Pr\{\gamma_{AR} \geq \gamma_{\text{th,P}}\} = Q_1\left(\sqrt{2\kappa_P}, \sqrt{\frac{2(1+\kappa_P)\gamma_{\text{th,P}}}{\bar{\gamma}_{AR}}}\right), \quad (22)$$

where $\bar{\gamma}_{AR} = P_A\beta_{AR}/\sigma_n^2$.

However, the success probability of the $R \rightarrow B$ link depends on the relay's transmit power P_R , which is given by (15) and is a function of $P_{S\text{-CSI}}$. Since the harvested energy E_R depends on the received RF power $P_{S\text{-CSI}}$, as defined in (10) and (11), the overall outage probability must be averaged over the PDF of $P_{S\text{-CSI}}$. An outage event occurs either when the received power at R falls below the sensitivity threshold $P_{\min}$, or when the instantaneous SNR at node B is lower than the threshold $\gamma_{\text{th,P}}$, given that $P_{S\text{-CSI}} \geq P_{\min}$. Accordingly, the total outage probability can be expressed as

$$\begin{aligned} \Pr\{\gamma_{RB} \geq \gamma_{\text{th,P}}\} &= \Pr\{P_{S\text{-CSI}} < P_{\min}\} \\ &+ \int_{P_{\min}}^{\infty} f_{P_{S\text{-CSI}}}(\xi) \\ &\times \Pr\{\gamma_{RB} \geq \gamma_{\text{th,P}} \mid P_{S\text{-CSI}} = \xi\} d\xi. \end{aligned} \quad (23)$$

The first term corresponds to the CDF $F_{P_{S\text{-CSI}}}(P_{\min})$, while the conditional success probability $\Pr\{\gamma_{RB} \geq \gamma_{\text{th,P}} \mid P_{S\text{-CSI}} = \xi\}$ can be obtained using (8). Thus,

$$\Pr\{\gamma_{RB} \geq \gamma_{\text{th,P}}\} = F_{P_{S\text{-CSI}}}(P_{\min}) + \int_{P_{\min}}^{\infty} f_{P_{S\text{-CSI}}}(\xi)$$

$$\times Q_1\left(\sqrt{2\kappa_P}, \sqrt{\frac{2(1+\kappa_P)\gamma_{\text{th,P}}}{\bar{\gamma}_{RB}(\xi)}}\right) d\xi, \quad (24)$$

with $\bar{\gamma}_{RB}(\xi) = P_R(\xi)\beta_{RB}/\sigma_n^2$.

By substituting (22) and (24) into (21), and subsequently substituting the resulting expression together with (20) into (19), the final analytical expression for the outage probability of the $A \rightarrow B$ link is obtained. Finally, let us remark that the analysis for the link originating from B follows by symmetry, leading to the same result.

B. SN PHASE OUTAGE PROBABILITY

During the SN Phase, the relay R transmits its own data to the gateway G. An outage occurs if the achievable rate falls below r_S . Assuming that the gateway G employs Maximum Ratio Combining (MRC) at the signals received in each antenna, and considering the phase duration $\delta_2 T$, an outage event corresponds to an instantaneous SNR γ_{RG} falling below the threshold $\gamma_{\text{th,RG}} = 2^{r_S/\delta_2} - 1$. Moreover, the instantaneous SNR at G, after MRC combining,⁶ is given by

$$\gamma_{RG} = \frac{P_{RG}}{\sigma_n^2} \sum_{m=1}^M |h_{RG,m}|^2. \quad (25)$$

Since the harvested energy E_R depends on the received RF power $P_{S\text{-CSI}}$, the overall outage probability must be averaged over the PDF of $P_{S\text{-CSI}}$. The total outage probability can then be written as

$$\begin{aligned} p_{\text{out,RG}} &= \Pr\{P_{S\text{-CSI}} < P_{\min}\} \\ &+ \int_{P_{\min}}^{\infty} \Pr\{\gamma_{RG} < \gamma_{\text{th,RG}} \mid P_{S\text{-CSI}} = \xi\} f_{P_{S\text{-CSI}}}(\xi) d\xi. \end{aligned} \quad (26)$$

The conditional probability inside the integral represents the CDF of $\sum_{m=1}^M |h_{RG,m}|^2$, which follows a non-central chi-squared distribution with $2M$ degrees of freedom, written as [43]

$$F_{\gamma_{RG}}(\gamma_{\text{th,RG}}) = 1 - Q_M\left(\sqrt{2\kappa_S M}, \sqrt{\frac{2(1+\kappa_S)\gamma_{\text{th,RG}}}{\bar{\gamma}_{RG}(\xi)}}\right), \quad (27)$$

where $Q_M(\cdot, \cdot)$ is the generalized Marcum Q -function of order M , and $\bar{\gamma}_{RG}(\xi) = P_{RG}(\xi)\beta_S/\sigma_n^2$ represents the average SNR, whose notation explicits the dependence of P_{RG} in (16) and (11) with the amount of received RF power $P_{S\text{-CSI}}$, which is the integration variable ξ in (23).

⁶The choice of S-CSI for energy beamforming in the WET phase prioritizes low overhead and steady energy delivery, making the low-complexity approach suitable for high-power transmission. Conversely, the secondary link requires MRC to compensate for the energy-constrained relay transmission. This requires Instantaneous CSI but is necessary to satisfy the reliability requirement, representing a deliberate engineering trade-off between complexity and performance. Furthermore, this model accounts for scenarios where the interval between WET and SN phases may exceed the channel coherence time, rendering instantaneous CSI-based energy beamforming impractical.

Consequently, with the aid of (27) the final expression can be written as

$$P_{\text{out, RG}} = F_{P_{\text{S-CSI}}}(P_{\min}) + \int_{P_{\min}}^{\infty} f_{P_{\text{S-CSI}}}(\xi) \times \left[1 - Q_M \left(\sqrt{2\kappa_S M}, \sqrt{\frac{2(1 + \kappa_S)\gamma_{\text{th, RG}}}{\bar{\gamma}_{\text{RG}}(\xi)}} \right) \right] d\xi. \quad (28)$$

IV. ASYMPTOTIC OUTAGE ANALYSIS

To provide deeper insights into the fundamental performance limitations of the system, we expand the analytical framework by deriving the asymptotic outage probabilities for both the high- and low-SNR regimes. In the high-SNR regime, the analysis demonstrates that the outage probability converges to an outage floor. This floor is fundamentally determined by the saturation limit (W) of the nonlinear EH circuit and the necessary fixed circuit power consumption (P_0), showing that system performance is not limited by transmit power but by EH capacity. Conversely, in the low-SNR Regime, the outage probability is dominated by the probability of energy insufficiency at the relay, which is primarily influenced by the sensitivity threshold ($P_{\min}$) and the efficiency of the energy beamforming. These asymptotic expressions clearly delineate the impact of design parameters across different operating conditions.

A. HIGH-SNR REGIME

The high-SNR regime corresponds to the scenario where the gateway's transmit power is very large ($P_G \rightarrow \infty$) or the distance between the gateway and the relay is very short ($d_{\text{GR}} \rightarrow 0$). Under these conditions, the received RF power at the relay, denoted by $P_{\text{S-CSI}}$, becomes significantly high.

According to the non-linear EH model described in (10), as the input power $P_{\text{S-CSI}} \rightarrow \infty$, the harvested DC power $\Phi(P_{\text{S-CSI}})$ asymptotically approaches its saturation level W . Consequently, the total harvested energy E_R in (11) converges to a maximum constant value, expressed as

$$E_{R, \max} = W\delta_1 T \left(1 - e^{-\frac{\delta_1 T}{\tau_c}} \right)^2. \quad (29)$$

This saturation implies that the relay's transmit powers, P_R and P_{RG} , also reach constant levels and cease to depend on the channel fading of the WET link. Their corresponding maximum values are given by

$$P_{R, \max} = \eta \left[\frac{3\mu E_{R, \max}}{(1 - \delta_1 - \delta_2)T} - P_0 \right], \quad (30)$$

and

$$P_{\text{RG}, \max} = \eta \left[\frac{(1 - \mu)E_{R, \max}}{\delta_2 T} - P_0 \right]. \quad (31)$$

With constant transmit powers, the integration over the PDF of $P_{\text{S-CSI}}$ is no longer required. In this case, the outage probabilities of the relay-dependent links converge to an *outage floor*, which is determined exclusively by the fading

characteristics of the PN or SN channels. Consequently, the asymptotic outage probability of the PN Phase is given by

$$P_{\text{out, RG}}^{\text{high}} \leq 1 - Q_M \left(\sqrt{2\kappa_S M}, \sqrt{\frac{2(1 + \kappa_S)\gamma_{\text{th, RG}}}{\bar{\gamma}_{\text{RG}, \max}}} \right), \quad (32)$$

with $\bar{\gamma}_{\text{RG}, \max} = P_{\text{RG}, \max} \beta_S / \sigma_n^2$.

Similarly, the success probability of the $R \rightarrow B$ link in (24) simplifies to

$$\Pr \{ \gamma_{\text{RB}} \geq \gamma_{\text{th, P}} \}^{\text{high}} \leq Q_1 \left(\sqrt{2\kappa_P}, \sqrt{\frac{2(1 + \kappa_P)\gamma_{\text{th, P}}}{\bar{\gamma}_{\text{RB}, \max}}} \right), \quad (33)$$

with $\bar{\gamma}_{\text{RB}, \max} = P_{R, \max} \beta_{\text{RB}} / \sigma_n^2$. The same happens for the $R \rightarrow A$ link.

The above results show that, even with an infinite amount of power available for energy harvesting, the overall system performance remains fundamentally constrained by the saturation limit of the EH circuit and the intrinsic fading characteristics of the communication channels.

B. LOW-SNR REGIME

The low-SNR regime occurs when the gateway's transmit power approaches zero ($P_G \rightarrow 0$) or when the distance between the gateway and the relay becomes very large ($d_{\text{GR}} \rightarrow \infty$). In this regime, the average received RF power at the relay is extremely low, and the predominant cause of outage is the high likelihood that the harvested energy is insufficient to sustain the operation of the relay's transceiver circuitry.

An outage inevitably occurs when the received power $P_{\text{S-CSI}}$ falls below the sensitivity threshold $P_{\min}$, or when the harvested energy E_R is insufficient to compensate for the fixed power consumption P_0 . Let $P_{\text{th, RG}}$ denote the minimum required incident RF power at the relay to guarantee $P_{\text{RG}} > 0$. Accordingly, an outage in the secondary link is certain whenever $P_{\text{S-CSI}} < P_{\text{th, RG}}$. Therefore, the outage probability can be lower-bounded by

$$P_{\text{out, RG}}^{\text{low}} \geq \Pr \{ P_{\text{S-CSI}} < P_{\text{th, RG}} \} = F_{P_{\text{S-CSI}}}(P_{\text{th, RG}}). \quad (34)$$

As the distance between the gateway and the relay increases ($d_{\text{GR}} \rightarrow \infty$), the path-loss term β_S in (5) approaches zero. Consequently, the parameter $\theta = \sqrt{2(1 + \kappa_S)/(P_G \beta_S)}$ in (9) grows without bound. In the CDF expression given in (8), a large value of the second argument of the Marcum- Q function drives the function towards zero. Therefore, for any threshold $\xi > 0$,

$$\lim_{d_{\text{GR}} \rightarrow \infty} F_{P_{\text{S-CSI}}}(\xi) = 1. \quad (35)$$

This confirms that the outage probabilities of both the secondary link, $p_{\text{out, RG}}$, and the relay-assisted primary path, $p_{\text{out, ARB}}^{\text{relay}}$, approach unity. In this asymptotic regime, the overall primary communication outage probability in (19) deteriorates to that of the direct transmission path only, i.e.,

$$\lim_{d_{\text{GR}} \rightarrow \infty} p_{\text{out, AB}} = p_{\text{out, AB}}^{\text{direct}}. \quad (36)$$

This result confirms that when the PB is located too far from the relay, the WET-enabled relay becomes ineffective, and the overall system performance depends solely on the direct communication link between the primary nodes.

V. NUMERICAL RESULTS

In this section, we present numerical results to validate the theoretical analysis and evaluate the outage performance of the proposed WET-enabled cooperative spectrum time-sharing network. All simulation-based results were generated using 5×10^6 independent channel realizations. The theoretical curves are obtained by the numerical integration of the final analytical outage expressions, which is necessary due to the complexity introduced by the nonlinear EH model and the requirement to average the conditional outage probability over the distribution of the harvested power.

Unless stated otherwise, the simulation setup uses the parameters listed in Table 3. Moreover, the total frame duration is in accordance with the typical time-on-air of some low power IoT networks, such as LoRaWAN and SigFox [46], [47], [48]. In addition, we clarify that the parameters adopted for the nonlinear EH model (including W , c_0 , and c_1) are consistent with the operating band and hardware. These specific values were experimentally derived from the Powercast P1110 evaluation board, which operates in the 902-928 MHz ISM band, ensuring an accurate representation of practical hardware performance at the simulated $f = 915$ MHz carrier frequency.

Fig. 3 depicts the outage probabilities of the primary and secondary communications as functions of the gateway-relay distance d_{GR} , including both theoretical and Monte Carlo simulation results, which agree very well, validating the derivations.⁷ Moreover, as d_{GR} increases, the relay-assisted links exhibit higher outage owing to the reduction in harvested energy E_R , which, in turn, lowers the relay transmit powers P_R and P_{RG} in PN phase 3 and in the SN phase, respectively. Among all, the secondary link $R \rightarrow G$ is able to achieve the lowest outage, benefiting from the receive diversity at the multi-antenna gateway. In contrast, the direct link $A \leftrightarrow B$ represents a benchmark scheme (e.g., a traditional system without spectrum-sharing, allocating $T/2$ to each phase) and remains constant, since it is independent of the relay. Overall, the cooperative transmissions $A \rightarrow B$ and $B \rightarrow A$ substantially outperform this direct-path benchmark when the gateway is located near the relay, underscoring the critical role of PB placement in enabling efficient energy transfer. Moreover, this result shows that in such WET-enabled spectrum-sharing network, the primary and the secondary nodes obtain mutual benefits compared to the benchmark. The primary communication becomes more reliable than in the case without relay assistance, while the secondary com-

TABLE 3. Simulation and System Parameters.

Parameter	Value	Ref.
Node Transmit Powers		
Gateway Transmit Power, P_G	40 dBm	[49]
Primary Node Transmit Power, P_A, P_B	10 dBm	[50]
Topology and Channel		
Node Distances, d_{AR}, d_{BR}	50 m	-
Node Distance, d_{AB}	100 m	-
Path Loss Exponent (Secondary Link), α_S	2	[51]
Path Loss Exponent (Primary Link), α_P	3	[14]
Rician K-factor (Secondary Link), κ_S	4 dB	[14]
Rician K-factor (Primary Link), κ_P	2 dB	[14]
Target data rate (Secondary Link), r_S	0.5 bits/s/Hz	-
Target data rate (Primary Link), r_P	1 bits/s/Hz	-
System and RF Parameters		
Carrier Frequency, f	915 MHz	[52]
Bandwidth, BW	100 kHz	[52]
Receiver Noise Figure	10 dB	[14]
Noise Power Spectral Density	-204 dBW/Hz	[14]
URA Antennas at G, M	9	-
Time Allocation		
Total Frame Duration, T	2000 ms	-
WET Phase Time Fraction, δ_1	0.4	-
SN Phase Time Fraction, δ_2	0.04	-
Relay/EH Parameters		
EH saturation level, W	10.73 mW	[44]
EH constant c_0	0.2308	[44]
EH constant c_1	5.365	[44]
Power Amplifier Efficiency, η	2/3	[53]
Fixed Circuitry Power, P_0	0.5 mW	[53]
Energy Harvesting Sensitivity, $P_{\min}$	-10 dBm	[44]
Capacitor Time Constant, τ_C	918 ms	[45]
Harvested Energy Fraction, μ	0.85	-

munication can achieve good QoS even without a dedicated spectrum.

Fig. 4 presents the outage probabilities of the $R \rightarrow G$ and $A \rightarrow B$ links, comparing Monte Carlo simulation results with the corresponding analytical and asymptotic expressions. As anticipated, the asymptotic curves closely follow the exact outage probabilities in both the high- and low-SNR regions, thereby confirming the accuracy and tightness of the proposed asymptotic analysis. Furthermore, Fig. 5 illustrates the outage probabilities of the $R \rightarrow G$ and cooperative $A \rightarrow B$ links as functions of the parameter δ_1 . As δ_1 increases, a larger portion of the frame is devoted to EH, resulting in greater harvested energy at the relay and, consequently, a lower outage probability for the $R \rightarrow G$ link. However, the extended WET duration reduces the time available for the PN phases 1–3, each occupying $\frac{(1-\delta_1-\delta_2)T}{3}$. This reduction in transmission time (and therefore increase in the transmitted data rate)

⁷Note that the SNR is inversely proportional to the distance d_{GR} , specifically given by $\text{SNR} \propto 1/(d_{GR})^{\alpha_{\text{wet}}}$. Therefore, a large d_{GR} corresponds to the low-SNR regime, while a small d_{GR} corresponds to the high-SNR regime.

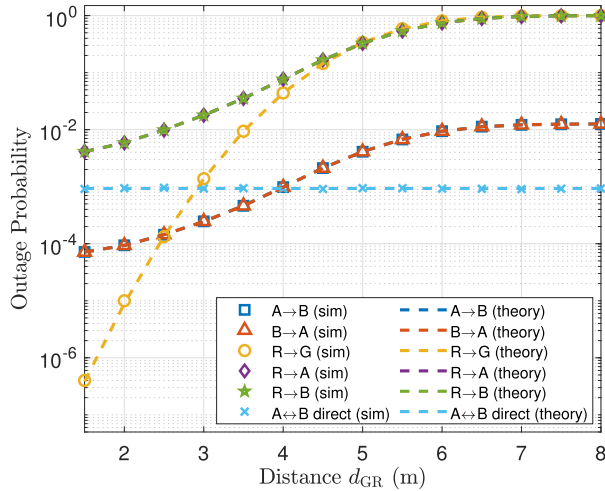


FIGURE 3. Outage probabilities of primary and secondary links versus gateway-relay distance d_{GR} .

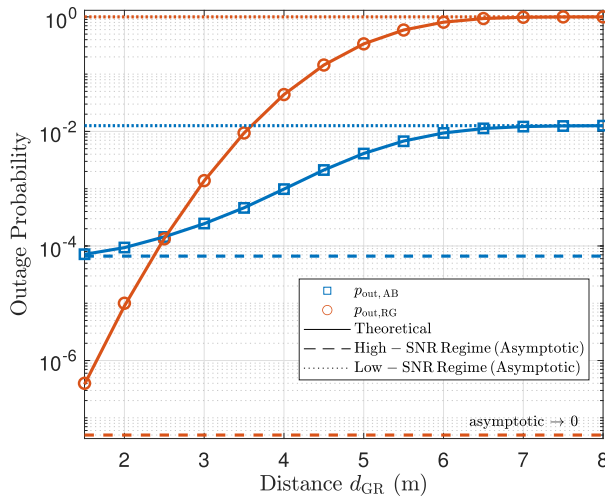


FIGURE 4. Outage probability of the secondary and primary communications versus d_{GR} , comparing Monte Carlo simulations, theoretical analysis, and asymptotic results.

increases the outage probability of the $A \rightarrow B$ link. Therefore, a clear trade-off arises: while a larger δ_1 enhances the energy availability and benefits the secondary relay-assisted link, it simultaneously deteriorates the performance of the primary transmission due to reduced data transmission intervals.

Fig. 6 analyzes the impact of the time fraction δ_2 . Interestingly, it is observed that the outage probabilities of both the $R \rightarrow G$ and $A \rightarrow B$ links increase with δ_2 . This behavior arises because a larger δ_2 reduces the time available for the cooperative PN phases among nodes A, B, and R, thereby increasing the effective target transmission rate required to maintain a constant throughput. Consequently, both primary and secondary communications experience performance degradation. Hence, δ_2 should be kept sufficiently small to preserve adequate time for cooperative relaying,

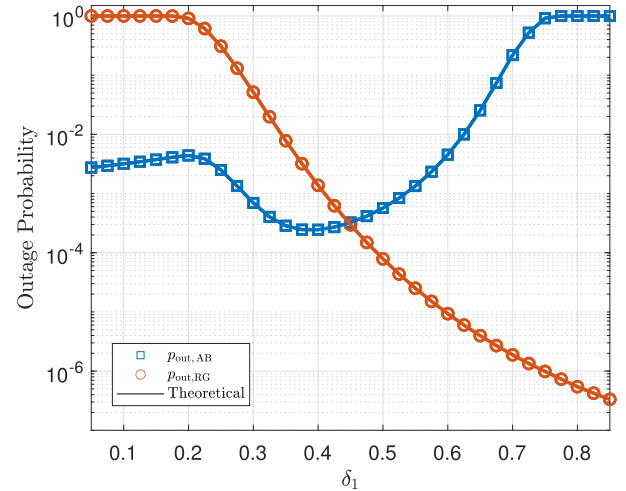


FIGURE 5. Outage probability of the secondary and primary communications versus δ_1 with fixed $\delta_2 = 0.04$ and $d_{GR} = 3$ m.

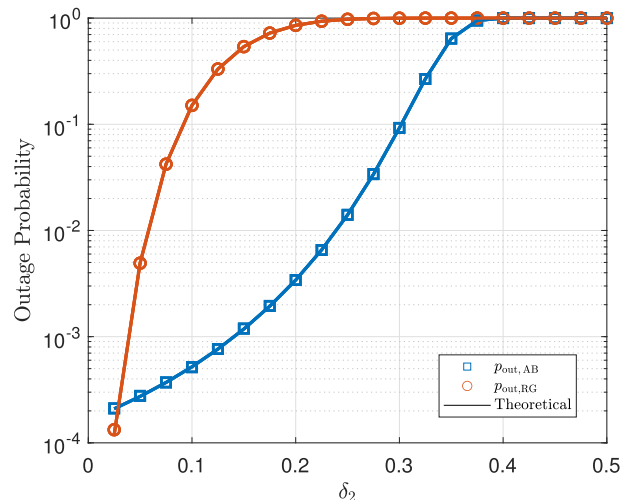


FIGURE 6. Outage probability of the secondary and primary communications versus δ_2 with fixed $\delta_1 = 0.4$ and $d_{GR} = 3$ m.

whereas δ_1 must be carefully optimized to guarantee efficient energy harvesting.

Next, Fig. 7 illustrates the outage probabilities versus the energy allocation factor μ . As μ increases, a greater portion of the harvested energy is allocated to primary forwarding, thereby enhancing the reliability of the $A \rightarrow B$ link and reducing its outage probability. In contrast, the $R \rightarrow G$ link suffers performance degradation since less energy remains available for the relay's own transmission. This opposite trend reveals the inherent energy-allocation trade-off between cooperative and secondary communications. Consequently, there exists an optimal value of μ that balances the performance of both primary and secondary links.

Fig. 8 shows the impact of the URA size M on the outage performance. Increasing M enhances beamforming gain and spatial diversity, leading to lower outage probabilities for both primary and secondary links. The secondary link experiences

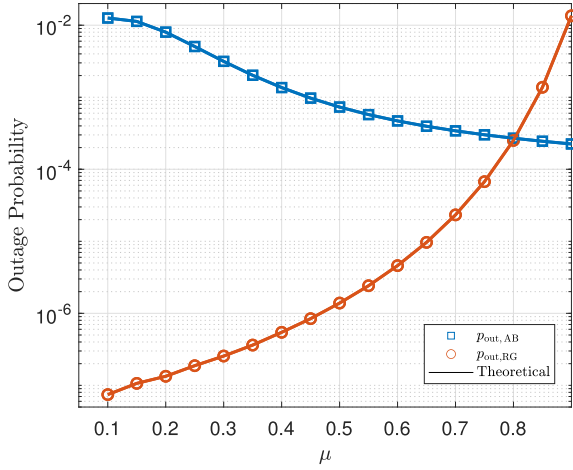


FIGURE 7. Outage probability of the secondary and primary communications versus μ with fixed $\delta_1 = 0.4$, $\delta_2 = 0.04$, and $d_{GR} = 3$ m.

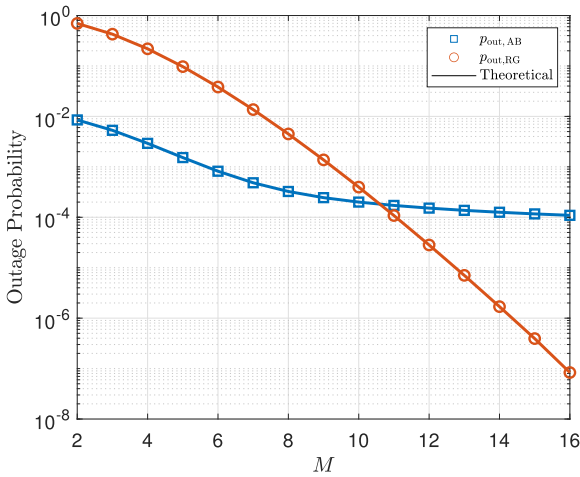


FIGURE 8. Outage probability of the secondary and primary links versus URA size M with fixed $\delta_1 = 0.4$, $\delta_2 = 0.04$, $\mu = 0.85$ and $d_{GR} = 3$ m.

a larger improvement due to stronger energy focusing toward the relay, while the primary link exhibits a smaller gain limited by residual interference and time allocation. The results indicate that enlarging the URA improves reliability, though the benefit saturates for large M .

Finally, to provide a holistic view of the system performance and capture the resource trade-offs discussed above, we define a unified weighted system outage probability as

$$p_{\text{out,sys}} = \omega p_{\text{out,AB}} + (1 - \omega) p_{\text{out,RG}}, \quad (37)$$

where $\omega \in [0, 1]$ represents the priority weight assigned to the PN. Fig. 9 illustrates $p_{\text{out,sys}}$ as a function of the time fraction δ_1 and the energy allocation factor μ for different priority weights. Fig. 9(a) reveals that the system outage exhibits convexity with respect to δ_1 . As ω increases (giving higher priority to the PN), the optimal δ_1 shifts to lower values to reserve more time for the PN phases. Conversely, Fig. 9(b) shows the impact of μ . When $\omega = 0.5$ (equal priority), a balanced μ minimizes the outage. However, as the priority

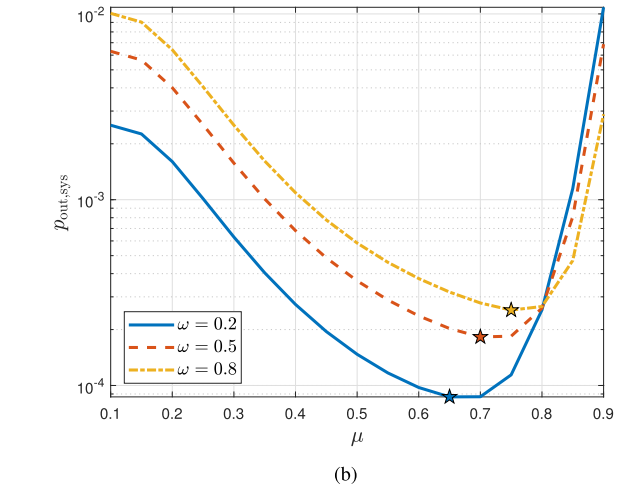
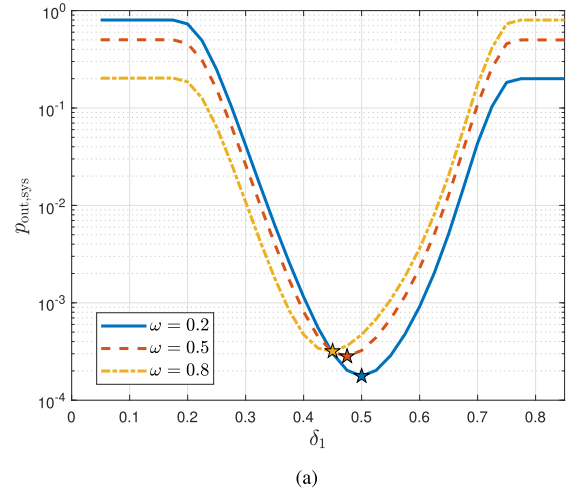


FIGURE 9. Weighted system outage probability ($p_{\text{out,sys}}$) for different priority weights ω : (a) versus δ_1 with $\mu = 0.85$; (b) versus μ with $\delta_1 = 0.4$. The convex shape confirms the existence of unique optimal resource allocations.

shifts toward the PN ($\omega = 0.8$), the optimal μ increases, allocating more harvested energy to the relaying phase. These results confirm that the derived analytical expressions provide a convex basis for optimizing system parameters to satisfy specific QoS priorities.

A. KEY PERFORMANCE INSIGHTS

The numerical results provide key insights into the performance and trade-offs of the proposed WET-enabled cooperative spectrum time-sharing network. First, the architecture achieves mutual reliability gains for both the primary (A $\rightarrow$ B) and secondary (R $\rightarrow$ G) links. The primary link benefits from the relay's assistance, and the secondary link achieves acceptable QoS opportunistically. This cooperative gain is fundamentally limited by d_{GR} , highlighting the critical role of relay positioning, as substantial benefits only occur when the gateway is near the relay.

Moreover, the WET time fraction (δ_1) presents a clear trade-off: a larger δ_1 enhances harvested energy (E_R) and benefits the secondary link, but simultaneously degrades the primary link's performance due to reduced data transmission time. Conversely, the relay's self-transmission time (δ_2) must be kept sufficiently small, as increasing δ_2 is detrimental to the outage probability of both links by limiting the cooperative phase duration.

The energy allocation factor (μ) also leads to an inherent trade-off. Increasing μ enhances the primary link's reliability ($A \rightarrow B$) but degrades the secondary link's performance ($R \rightarrow G$); thus, an optimal μ is necessary to balance both systems' QoS requirements. Finally, the size of the URA at the gateway is a key factor, as increasing M significantly improves reliability for both communications by enhancing energy beamforming gain and receive spatial diversity.

These opposing effects, particularly those governed by δ_1 and μ , highlight a critical resource trade-off between the PN and SN. Therefore, our detailed analysis of the individual outage probabilities is foundational, enabling the careful optimization of these system parameters to achieve a desired balance or to maximize a weighted combined performance metric.

VI. CONCLUSION

This paper presented a performance analysis of an energy-harvesting cooperative spectrum time-sharing radio network operating under statistical channel knowledge and a realistic nonlinear rectenna model. Expressions were derived for the outage probabilities of both primary and secondary links under Rician fading. The results showed that nonlinear energy conversion introduces outage floors at high-SNR, while energy insufficiency dominates in the low-SNR regime. Furthermore, the performance is strongly influenced by the power allocation ratio, beamforming accuracy, and the distance between the power beacon and the relay.

Numerical results validated the analytical derivations and highlighted design trade-offs between energy efficiency and link reliability. These findings offer valuable insights for the development of cooperative spectrum-sharing IoT networks where energy harvesting and spectrum sharing coexist. Moreover, our results show that both the primary and the secondary network can benefit, illustrating the potential of the proposed strategy for next generation IoT. Future work may include other aspects such as the analysis of multi-relay selection strategies or the integration of intelligent reflecting surfaces to further enhance energy transfer efficiency. Additionally, to fully align the receiver processing with the theme of complexity reduction, an important extension is the analysis of blind or semi-blind detection scenarios for the primary network links.

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